GWA CALCULATOR

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PROBLEM STATEMENT/RATIONALE

Most Pisay scholars find it difficult to manage and assess their academic performance due to how complex the grading system is, along with the possible pressure that students can get with maintaining high grades. Considering each subject has its own weights on different assessments, this makes it very tedious for scholars to find out which components affects their final grade significantly. Because of this, a Pisay-based GWA computation system was made that can provide early, accurate, and reliable grades based on the students' assessment scores.

PROJECT OVERVIEW

Our project allows users to:

-Calculate their GWA of 1 subject (up to the entire school year)

-Calculate their GWA for all subjects for 1 quarter

PROJECT RELEVANCE

Our project is relevant to:

-G8 Students who need to check in their GWA at the end/middle of the quarter

-Teachers who need a fast way to keep on calculating grades for students

OBJECTIVES

- To help Pisay scholars estimate and calculate their grades in advance before they are officially released, in order to keep them prepared for the possible outcome.
- To allow students to calculate their GWA for one quarter.
- To help students understand how different percentages of their assessment scores affect their grades.
- To provide a simple and user-friendly tool for grade computation in higher year levels.
- To reduce uncertainty and stress that students may have related to their academic performance.

TECHNICAL APPROACH

Reuirements Gathering

-Mapping the PSHS 2025-2026 S.Y. grading weights

Algorithm Design

-Designning the Python logic for cumulative and non-cumulative quarters

Database Intergration

-Establishing the JSON paths for data storage

Error Handling

-Implementation of “try-except” and “while” blocks to catch invalid or mismatched units

CODE: LOGIN/SIGNUP

```python
#Login/signup
login_option=int(input("""Hello, user.
Input 1 to login(Note for login the users and corresponding passwords are user1: pass1 and user2:pass2)
Input 2 to sign up
: """))
print("")

#Checking user and password
import time
password_att=0
if login_option==1:
    username=input("Enter username: ")
    while username not in users:
        username=input("User not found, try again: ")
    password=input("Enter password: ")
    while users[username]!=password:
        password_att=password_att+1
        password=input("Wrong password, try again: ")
        #Account lock
        if password_att>=3:
            print("Try again after 10s")
            password_att=0
            time.sleep(10)
    print(f"Welcome, {username}!")

#Sign-in
if login_option==2:
    new_user=input("Enter new username: ")
    new_password=input("Enter new password: ")
    #Confirm new user
    users.update({new_user:new_password})
    username=input("Confirm username: ")
    while username not in users:
        username=input("User not found, try again: ")
    password=input("Confirm password: ")
    while users[new_user]!=password:
        password_att=password_att+1
        password=input("Wrong password, try again: ")
        #Account lock
        if password_att>=3:
            print("Try again after 10s")
            password_att=0
            time.sleep(10)
    print(f"Welcome, {username}!")
```

INPUTS/OUTPUTS

LOGIN SYSTEM

```
Hello, user.
Input 1 to login(Note for login the users and corresponding passwords are user1: pass1 and user2:pass2)
Input 2 to sign up
: 1

Enter username: user1
Enter password: pass1
Welcome, user1!
Welcome to the Grade 8 Pisay GWA Calculator
Would you like to
1. Calculate your grade per subject
2. Calculate your GWA for the entire quarter
Or click any other number to exit
Please enter your response: |
```

INPUTS/OUTPUTS

SIGNUP SYSTEM

```
Hello, user.
Input 1 to login(Note for login the users and corresponding passwords are user1: pass1 and user2:pass2)
Input 2 to sign up
: 2

Enter new username: CJ
Enter new password: Ilagan
```

CODE: GWA Per Subject

```python
elif subjectnum==7:
    subject="Physics 1"
    #FA calculation
    while answer!=2:
        print("Please input your score for the FA")
        fascore=float(input("Score: "))
        print("Please type your maximum score for the FA")
        famaxscore=float(input("Max Score: "))
        while famaxscore<fascore:
            print("Your score can not be greater than the max score")
            fascore=float(input("Please input your score for the FA: "))
            famaxscore=float(input("Please type your maximum score for the FA: "))
        answer=int(input("Do you still have any fas to input(please input 1 or 2, 1 for yes 2 for no): "))
        while answer!=1 and answer!=2:
            print("Please pick between 1 or 2")
            answer=int(input("Do you have anymore fas to input: "))
        totalfascore=totalfascore+fascore
        totalfamaxscore=totalfamaxscore+famaxscore
    #AA calculation
    while answer2!=2:
        print("Please input your score for the AA")
        aascore=float(input("Score: "))
        print("Please type your maximum score for the AA")
        aamaxscore=float(input("Max Score: "))
        while aamaxscore<aascore:
            print("Your score can not be greater than the max score")
            aascore=float(input("Please input your score for the AA: "))
            aamaxscore=float(input("Please type your maximum score for the AA: "))
        answer2=int(input("Do you still have any aas to input(please input 1 or 2, 1 for yes 2 for no): "))
        while answer2!=1 and answer2!=2:
            print("Please pick between 1 or 2")
            answer2=int(input("Do you have anymore aas to input: "))
        totalaascore=totalaascore+aascore
        totalaamaxscore=totalaamaxscore+aamaxscore
    #LT calculation
    while answer3!=2:
        print("Please input your score for the LT")
        ltscore=float(input("Score: "))
        print("Please type your maximum score for the LT")
        ltmaxscore=float(input("Max Score: "))
        while ltmaxscore<ltscore:
            print("Your score can not be greater than the max score")
            ltscore=float(input("Please input your score for the LT: "))
            ltmaxscore=float(input("Please type your maximum score for the LT: "))
        answer3=int(input("Do you still have any lts to input(please input 1 or 2, 1 for yes 2 for no): "))
        while answer3!=1 and answer3!=2:
            print("Please pick between 1 or 2")
            answer3=int(input("Do you have anymore lts to input: "))
        totalltscore=totalltscore+ltscore
        totalltmaxscore=totalltmaxscore+ltmaxscore
    #Raw grade calculation
    if totalfamaxscore == 0:
        fapercentage = 0
    else:
        fapercentage = totalfascore/totalfamaxscore*0.30

    if totalaamaxscore == 0:
        aapercentage = 0
    else:
        aapercentage = totalaascore/totalaamaxscore *0.15

    if totalltmaxscore == 0:
        ltpercentage = 0
    else:
      ltpercentage=totalltscore/totalltmaxscore*0.55

    GWA=fapercentage+aapercentage+ltpercentage
```

```python
    else:
      ltpercentage=totalltscore/totalltmaxscore*0.55

    GWA=fapercentage+aapercentage+ltpercentage
    grading=int(input("Please input 1 for first grading and please input 2 for the other gradings: "))

    while grading!=1 and grading!=2:
        grading=int(input("Please input 1 or 2: "))
    #Raw grade calculation with previous grade
    if grading==2:
        previousrawgrade=float(input(f"Please input your raw grade(0-100) in {subject}: "))
        while previousrawgrade>100 or previousrawgrade<0:
            previousrawgrade=float(input("Please input your raw grade again it can not over 100 or under 0: "))
        GWA=(GWA*2/3) + (previousrawgrade/300)
    rawgrade=round(GWA*100,2)

    print(f"Raw grade: {rawgrade}")
    #GWA calculation
    if GWA<=1.00 and GWA>=0.95:
        print(f"Your grade for {subject} is 1.00")
    elif GWA<0.95 and GWA>=0.9:
        print(f"Your grade for {subject} is 1.25")
    elif GWA<0.9 and GWA>=0.84:
        print(f"Your grade for {subject} is 1.50")
    elif GWA<0.84 and GWA>=0.78:
        print(f"Your grade for {subject} is 1.75")
    elif GWA<0.78 and GWA>=0.72:
        print(f"Your grade for {subject} is 2.00")
    elif GWA<0.72 and GWA>=0.66:
        print(f"Your grade for {subject} is 2.25")
    elif GWA<0.66 and GWA>=0.6:
        print(f"Your grade for {subject} is 2.50")
    elif GWA<0.6 and GWA>=0.55:
        print(f"Your grade for {subject} is 2.75")
    elif GWA<0.55 and GWA>=0.5:
        print(f"Your grade for {subject} is 3.00")
    elif GWA<0.5 and GWA>=0.4:
        print(f"Your grade for {subject} is 4.00")
    elif GWA<0.4 and GWA>=0:
        print(f"Your grade for {subject} is 5.00")
    else:
        print("Invalid Grade")
    break
```

```python
response=int(input("Would you like to try for another subject or quit the program(please enter 1 or 2): "))
while response!=1 and response!=2:
    response=int(input("Please enter 1 or 2: "))
```

INPUTS/OUTPUTS

1.) GWA PER SUBJECT

```
Please enter your response: 1
which subject would you like to calculate your GWA?
1. Math 3
2. Math 2
3. ADTECH 2
4. Social Science 2
5. Biology 1
6. Chemistry 1
7. Physics 1
8. Earth Science 1
9. PEHM 2
10. Values Education 2
11. Computer Science 2
12. English 2
13. Filipino 2
Please input the subject with its corresponding number:
```

```
Please input the subject with its corresponding number: 7
Please input your score for the FA
Score: 65
Please type your maximum score for the FA
Max Score: 12
Your score can not be greater than the max score
Please input your score for the FA: 65
Please type your maximum score for the FA: 67.25
Do you still have any fas to input(please input 1 or 2, 1 for yes 2 for no): 2
Please input your score for the AA
Score: 64.3
Please type your maximum score for the AA
Max Score: 64.345
Do you still have any aas to input(please input 1 or 2, 1 for yes 2 for no): 1
Please input your score for the AA
Score: 50
Please type your maximum score for the AA
Max Score: 50
Do you still have any aas to input(please input 1 or 2, 1 for yes 2 for no): 2
Please input your score for the LT
Score: 47
Please type your maximum score for the LT
Max Score: 50
Do you still have any lts to input(please input 1 or 2, 1 for yes 2 for no): 1
Please input your score for the LT
Score: 29
Please type your maximum score for the LT
Max Score: 30
```

INPUTS/OUTPUTS

1.1) GWA PER SUBJECT FOR 1ST QUARTER

```
Please input your score for the LT: 42
Please type your maximum score for the LT: 42
Do you still have any lts to input(please input 1 or 2), 1 for yes 2 for no: 2
Please input your score for the perio: 50
Please input the max score for the perio: 50
Please input 1 for first grading and please input 2 for the other gradings: 1
Raw grade: 95.51
Your grade for Math 3 is 1.25
Would you like to try for another subject or quit the program(please enter 1 or 2):
```

INPUTS/OUTPUTS

1.2) GWA PER SUBJECT FOR 2ND-4TH QUARTERS

```
Please input the subject with its corresponding number: 1
Please input your score for the FA
Score: 50
Please type your maximum score for the FA
Max Score: 50
Do you still have any fas to input(please input 1 or 2), 1 for yes 2 for no: 2
Please input your score for the AA
Score: 50
Please type your maximum score for the AA
Max Score: 50
Do you still have any aas to input(please input 1 or 2), 1 for yes 2 for no: 2
Please input your score for the LT
Score: 50
Please type your maximum score for the LT
Max Score: 50
Do you still have any lts to input(please input 1 or 2), 1 for yes 2 for no: 2
Please input your score for the perio: 50
Please input the max score for the perio: 50
Please input 1 for first grading and please input 2 for the other gradings: 2
Please input your raw grade(0-100) in Math 3: 96.45
Raw grade: 98.82
Your grade for Math 3 is 1.00
Would you like to try for another subject or quit the program(please enter 1 or 2):
```

CODE:

OPTION 2:

```python
elif response1==2:
    subjects=["Math 3","Math 2","ADTECH","SocSci","Biology","Chemistry","Physics","Earth Science","PEHM","Filipino","Computer Science","English","Values Education"]
    grades = [0] * 12
    gradesformatted=[]
    valid_grades = [1.00, 1.25, 1.50, 1.75, 2.00, 2.25, 2.50, 2.75, 3.00, 4.00, 5.00]
    print("Please input your grades and only pick from 1.00,1.25,1.50,1.75,2.00,2.25,2.50,2.75,3.00,4.00, and 5.00 unless its Values Education")
    g = 0
    for i in range(13):
        print(f"{subjects[i]}:")
        subject=subjects[i]
        if subject==subjects[12]:
            status=input("Please input complete or incomplete:")
            while status!="complete" and status!="incomplete":
                status=input("Please input only complete or incomplete: ")
        else:
            grade=float(input("Please input your grade: "))
            while grade not in valid_grades:
                print("Invalid grade please input your grade again")
                grade=float(input(f"{subjects[i]}: "))
            grades[g]=round(grade,2)
            g += 1
    if status=="complete":
        print(f"""{subjects[0]}: {grades[0]:.2f}
              {subjects[1]}: {grades[1]:.2f}
              {subjects[2]}: {grades[2]:.2f}
              {subjects[3]}: {grades[3]:.2f}
              {subjects[4]}: {grades[4]:.2f}
              {subjects[5]}: {grades[5]:.2f}
              {subjects[6]}: {grades[6]:.2f}
              {subjects[7]}: {grades[7]:.2f}
              {subjects[8]}: {grades[8]:.2f}
              {subjects[9]}: {grades[9]:.2f}
              {subjects[10]}: {grades[10]:.2f}
              {subjects[11]}: {grades[11]:.2f}
              {subjects[12]}: {status}""")
        TotalGWA=sum(grades)/len(grades)
        TotalGWA=round(TotalGWA,2)
        if TotalGWA<=1.50:
            print(f"Final GWA: {TotalGWA}, Congrats you are apart of the Director's List")
        else:
            print(f"Final GWA: {TotalGWA}")

    else:
        print(f"""{subjects[0]}: {grades[0]:.2f}
              {subjects[1]}: {grades[1]:.2f}
              {subjects[2]}: {grades[2]:.2f}
              {subjects[3]}: {grades[3]:.2f}
              {subjects[4]}: {grades[4]:.2f}
              {subjects[5]}: {grades[5]:.2f}
              {subjects[6]}: {grades[6]:.2f}
              {subjects[7]}: {grades[7]:.2f}
              {subjects[8]}: {grades[8]:.2f}
              {subjects[9]}: {grades[9]:.2f}
              {subjects[10]}: {grades[10]:.2f}
              {subjects[11]}: {grades[11]:.2f}
              {subjects[12]}: {status}""")
        TotalGWA=sum(grades)/len(grades)
        TotalGWA=round(TotalGWA,2)
        print(f"Final GWA: {TotalGWA}(Probationary Status: Please complete your VE requirements)")
```

INPUTS/OUTPUTS

2.) ENTIRE GWA PER QUARTER

```
Please input your grades and only pick from 1.00,1.25,1.50,1.75,2.00,2.25,2.50,2.75,3.00,4.00, and 5.00 unless its Value
s Education
Math 3:
Please input your grade: 1.00
Math 2:
Please input your grade: 1.00
ADTECH:
Please input your grade: 1.25
SocSci:
Please input your grade: 1.25
Biology:
Please input your grade: 1.50
Chemistry:
Please input your grade: 1.67
Invalid grade please input your grade again
Chemistry: 1.25
Physics:
Please input your grade: 0
Invalid grade please input your grade again
Physics: 1.00
Earth Science:
Please input your grade: 1.00
PEHM:
Please input your grade: 1.00
Filipino:
Please input your grade: 1.00
Computer Science:
Please input your grade: 1.00
English:
Please input your grade: 1.00
```

```
English:
Please input your grade: 1.00
Values Education:
Please input complete or incomplete:complete
Math 3: 1.00
              Math 2: 1.00
              ADTECH: 1.25
              SocSci: 1.25
              Biology: 1.50
              Chemistry: 1.25
              Physics: 1.00
              Earth Science: 1.00
              PEHM: 1.00
              Filipino: 1.00
              Computer Science: 1.00
              English: 1.00
              Values Education: complete
Final GWA: 1.1, Congrats you are apart of the Director's List
Thank you for using our program
The program is closing, Press anything to exit
```